Dataset Description: I chose this dataset as it goes over player transfer value over time, which would allow me to create a candlestick chart. It also aligns with my interests and relates to the type of analysis and data visualization I would like to work on in the future. This dataset includes two files that are relevant for two different charts. One is player values, which is be useful for a bubble chart, the other file is player value history, which is useful for candlestick chart. The data is collected from a reputable source Transfermarkt Source: https://www.kaggle.com/datasets/kanchana1990/football-transfer-value-intelligence-2024

In [59]:
```
# import libraries
import plotly.express as px
import pandas as pd
import plotly.graph_objects as go
```

In [60]:
```
value_history = pd.read_csv('transfermarkt_value_history.csv')
```

In [61]:
```
value_history.head()
```

Out[61]:

	player_id	valuation_date	value_eur	value_display	club	age_at_date	year	month
0	108390	01/06/2016	35000000	€35.00m	Chelsea FC	24	2016	5
1	108390	01/07/2015	35000000	€35.00m	Chelsea FC	23	2015	6
2	108390	01/08/2012	15000000	€15.00m	Atlético de Madrid	20	2012	7
3	108390	01/08/2016	35000000	€35.00m	Chelsea FC	24	2016	7
4	108390	02/01/2018	50000000	€50.00m	Chelsea FC	25	2018	1

In [62]:
```
player_values = pd.read_csv('transfermarkt_player_values.csv')
```

In [63]:
```
player_values.head()
```

Out[63]:

	player_id	name	age	nationality	position	position_group	current_club	league_name	current_value_eur	current_value_tier	...	mean_yoy_growth_rate	num_valuation_points	num_clubs_career	trajectory	is_at_peak	ever_100m	ever_50m	ev
0	418560	Erling Haaland	25.0	Norway	€200.00m	Other	Norway	Premier League	200000000.0	World Class (100M+)	...	1.6329	26	5	growing	0	1	1	
1	433177	Bukayo Saka	24.0	England	€130.00m	Other	England	Premier League	130000000.0	World Class (100M+)	...	0.6972	22	1	growing	0	1	1	
2	357662	Declan Rice	27.0	England	€120.00m	Other	England	Premier League	120000000.0	World Class (100M+)	...	1.6100	24	2	growing	0	1	1	
3	568177	Cole Palmer	23.0	England	€120.00m	Other	England	Premier League	120000000.0	World Class (100M+)	...	2.0823	15	3	growing	0	1	1	
4	349066	Alexander Isak	26.0	Sweden	€120.00m	Other	Sweden	Premier League	120000000.0	World Class (100M+)	...	1.5154	28	6	growing	0	1	1	

5 rows × 36 columns

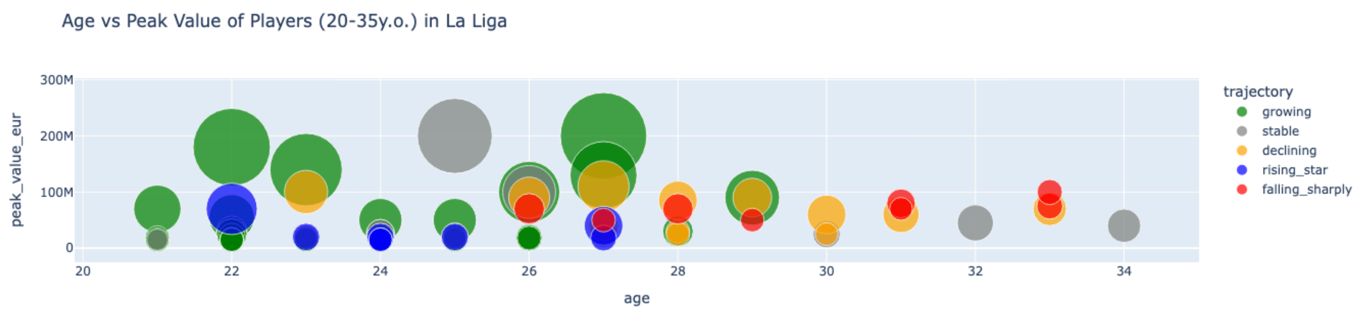
Part 1: Bubble Chart Create a Bubble Chart: Utilize your dataset to design a bubble chart that illustrates a specific data trend or relationship. Ensure your bubble chart is clear and informative by:

Labeling axes clearly. Choosing appropriate scales. Using colors or annotations to emphasize key points. Analysis: Discuss your rationale for creating the bubble chart, your observations, and any interesting conclusions you can draw from the data.

In [64]:
```
print(player_values.columns.tolist())
#displays all the columns in the dataset
```

```
['player_id', 'name', 'age', 'nationality', 'position', 'position_group', 'current_club', 'league_name', 'current_value_eur', 'current_value_tier', 'peak_value_eur', 'peak_value_tier', 'peak_date', 'peak_club', 'age_at_peak', 'first_value_eur', 'first_date', 'last_value_eur', 'last_date', 'career_span_years', 'years_to_peak', 'value_cagr', 'value_to_peak_cagr', 'value_multiplier_x', 'post_peak_decline_pct', 'value_volatility', 'mean_yoy_growth_rate', 'num_valuation_points', 'num_clubs_career', 'trajectory', 'is_at_peak', 'ever_100m', 'ever_50m', 'ever_10m', 'data_source', 'dataset_built_at']
```

In [65]:
```
bubble = px.scatter(player_values.query('league_name=="La Liga" and 20<age<35'), x='age', y='peak_value_eur',
                    size='current_value_eur', color='trajectory',
                    hover_name='name', size_max=60,
                    title='Age vs Peak Value of Players (20-35y.o.) in La Liga',
                   color_discrete_map = {
                       'growing':'green',
                       'stable':'grey',
                       'rising_star':'blue',
                       'declining':'orange',
                       'falling_sharply':'red'})
bubble.show()
#creates a bubble chart
```



Analysis: This bubble chart shows players less than 35 years old and their transfer value. The size of the bubble represents a player's current market value. Based on this graph, a club can make a decision whether it is worth it to buy a player from the other club. For example, a big grey circle in the 25y.o. range represents Vinicius Jr, a player whose maket value has stopped growing. Based on this information, it becomes clear that the player has stopped improving, and his transfer value reflects that. Furthermore, this graph shows that after the age of 29, the players stop improving in the market value category. The only exception here is Raphinha(29, $80M). This could be explained by his recent peak form and performances during the matches. Overall, the chart confirms that the most optimal time to purchase a player is when he is between 20-24 years of age in order to get the best return of investment.

In [66]:
```
#-------------------------------------------------------------------------------------------------------------------------------------
```

Part 2: Funnel Chart Create a Funnel Chart: Design a funnel chart to represent stages in a process or to highlight a data flow using your chosen dataset. Make your funnel chart effective by:

Clearly labeling axes. Selecting suitable scales. Applying colors or annotations to highlight important aspects. Analysis: Explain why you opted for a funnel chart, describe your findings, and detail the conclusions you reached based on the data.

In [67]:
```
print(player_values['current_value_tier'].unique())
#displays different categories in the current value tier column
```

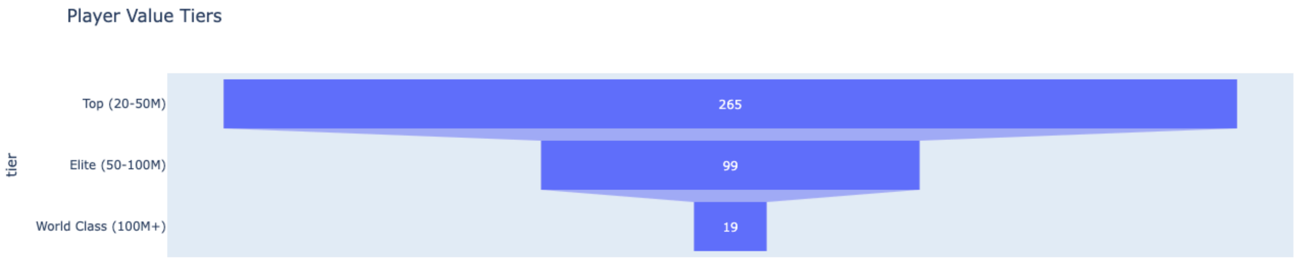
```
<StringArray>
['World Class (100M+)',     'Elite (50–100M)',        'Top (20–50M)',
     'Regular (5–20M)',    'Developing (<5M)']
Length: 5, dtype: str
```

In [68]:
```
tiers = ['Top (20–50M)', 'Elite (50–100M)', 'World Class (100M+)']
#creates a list with all players in each tier
tier_numbers = player_values['current_value_tier'].value_counts().reindex(tiers).reset_index()
#reorders results to fit into the tiers list
tier_numbers.columns = ['tier','count']
#creates a dataframe with separated tier and player numbers
tier_numbers
```

Out[68]:

	tier	count
0	Top (20–50M)	265
1	Elite (50–100M)	99
2	World Class (100M+)	19

In [69]:
```
funnel = px.funnel(tier_numbers, x='count', y='tier', title = 'Player Value Tiers')
funnel.show()
#creates a funnel chart
```



Analysis: This funnel chart shows the distribution of players in each tier. By looking at this graph, it becomes clear that the majority of players in the top 5 european leages are valued between 20 and 50 million. Furthermore, reaching the 'world class' tier is extremely rare -- out of the player in the top tier, only 7% make it to the 'world class' tier. This demonstrates how difficult it is to reach this status -- a player needs to have exceptional abilities in order to be valued this highly.

In [70]:
```
world_class_tier = player_values.query('current_value_tier == "World Class (100M+)"')
print(world_class_tier[['name','current_value_eur','league_name']])
#creates a table showing only world class players and the league the play in
```

```
                  name  current_value_eur     league_name
0       Erling Haaland        200000000.0  Premier League
1          Bukayo Saka        130000000.0  Premier League
2          Declan Rice        120000000.0  Premier League
3          Cole Palmer        120000000.0  Premier League
4       Alexander Isak        120000000.0  Premier League
5       Moisés Caicedo        110000000.0  Premier League
6        Florian Wirtz        110000000.0  Premier League
103      Kylian Mbappé        200000000.0         La Liga
104       Lamine Yamal        200000000.0         La Liga
105    Jude Bellingham        160000000.0         La Liga
106    Vinicius Junior        150000000.0         La Liga
107              Pedri        140000000.0         La Liga
108  Federico Valverde        120000000.0         La Liga
109      Julián Alvarez        100000000.0         La Liga
207      Jamal Musiala        130000000.0      Bundesliga
208      Michael Olise        130000000.0      Bundesliga
408            Vitinha        110000000.0         Ligue 1
409         João Neves        110000000.0         Ligue 1
410    Ousmane Dembélé        100000000.0         Ligue 1
```

This table represents all world class (above 100M) players in top 5 leagues. From this table, it is clear that the most valued players are in the English Premier League, followed by La Liga, Bundesliga, and Ligue 1. However, Serie A, an italian league, has no players in the world class tier. This shows how italian football has declined in the recent years. From having such notable players as Kaka, Ronaldo (R9), Pogba, Cristiano Ronaldo, and so on to now not having any is a worrying sign for italian football.

In [71]:
```
#--------------------------------------------------------------------------------------------------------------------------------
```

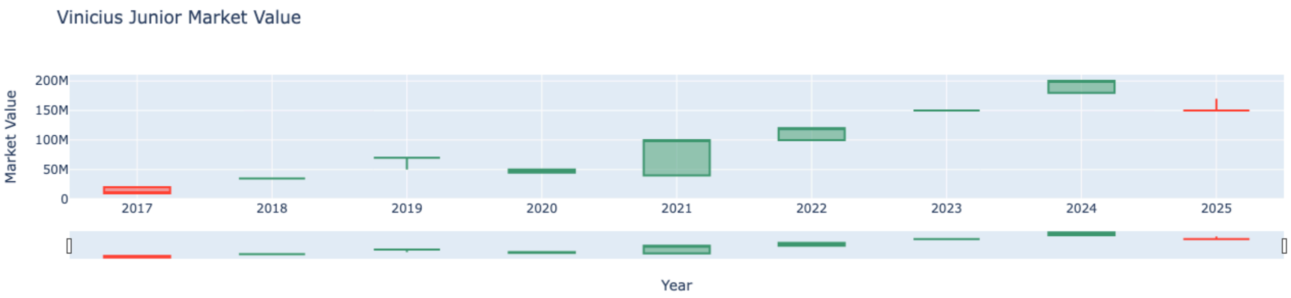
Part 3: Candlestick Chart Create a Candlestick Chart: Use the provided stock market dataset or a dataset of your choice to construct a candlestick chart, which is commonly used to show price movements over time. Your candlestick chart should be well-designed, featuring:

Clear axis labels. Appropriate scaling. Colors or annotations to underscore significant points. Analysis: Justify your decision to use a candlestick chart, discuss your observations, and articulate the insights you gained from the data.

In [76]:
```
player = 'Vinicius Junior'
player_id = player_values[player_values['name'] == player]['player_id'].values[0]
#finds the player id in the player_values dataframe that belongs to Vinicius Junior
player_history = value_history[value_history['player_id'] == player_id].copy()
#finds the valuation history of the player and creates a copy
open_values = player_history.groupby('year')['value_eur'].first()
close_values = player_history.groupby('year')['value_eur'].last()
high_values = player_history.groupby('year')['value_eur'].max()
low_values = player_history.groupby('year')['value_eur'].min()
#groups rows by year by only taking the value_eur column
```

In [80]:
```
candle = go.Figure(data=[go.Candlestick(
    x = open_values.index,
    open = open_values.values,
    high = high_values.values,
    low = low_values.values,
    close = close_values.values)])
candle.update_layout(
    title = dict(text='Vinicius Junior Market Value'),
    yaxis = dict(title=dict(text='Market Value')),
```

```
    xaxis = dict(title=dict(text='Year')))
candle.show()
#creates a candlestick chart
```



Analysis: The candlestick chart demonstrates the market value of Vinicius Junior. By looking at this chart, several conclusions can be made. First, in 2017, his transfer value went down by 10M. This was his first season in senior football in the Brazilian League. This drop in price can be justified by the different level of football that the player has been introduced to and could not match. Nonetheless, after being purchased by Real Madrid, his market value tripled, going from 10M to 35M. This jump can be explained solely by the reputation of the club, as the player himself did not get many minutes on the pitch to fairly justify this increase in price. The drop of 2020 can be explained by the drop in the entire football market due to COVID19. The biggest jump in player market value can be seen in 2021. During that season, Vinicius Junior and Real Madrid have been on the top of the European Football by winning La Liga and Champions League, where the player scored the only goal in the final. Throughout that season, the player had demonstrated incredible commitment and a massive jump in quality of play, which reflected on the increase in market value. Continuing up until 2025 season, Junior's value has been consistently increasing. However, with the sudden drop in the quality of performances during the last season, his valuation dropped as well from 200M to 150M. Overall, the market value of Vinicius Junior resembles that of a high-growth stock -- rapidly increasing in the early years and now stabilizing at the elite level.

In [ ]: